

Social Exclusion and Social Preferences: Evidence from Colombia's Leper Colony[†]

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This paper explores the intergenerational consequences of social exclusion on prosociality. A lab-in-the-field approach in the historical region of Colombia's leper colony reveals that descendants of socially excluded individuals are locally altruistic and extend such altruism to outsiders who have undergone similar circumstances. These individuals also display mistrust toward those who have, historically, been exclusionary—in this case, doctors. The content of historical narratives shared by ancestors who were excluded, which emphasize the endured mistreatment and doctors' historical misinformation, is one mechanism that partially explains the intergenerational patterns. (JEL D64, H51, I12, I18, N36, N96, Z13)

The social exclusion of minorities is historically widespread and remains a prevalent phenomenon in both developing and developed regions. Moreover, the number of groups and individuals subjected to social exclusion will likely increase in the coming years and decades (United Nations, Department of Economic and Social Affairs 2016). These socially excluded communities are unable to enjoy opportunities available to the majority of the population and are, indeed, subject to traumas deriving from their stigmatization and mistreatment. Despite the relevance of this phenomenon, little is known about its lasting implications on attitudes and behaviors. Can the experiences of social exclusion affect beliefs and behaviors across generations even when descendants are no longer subject to the adverse conditions endured by their ancestors?

*Department of Economic, Dartmouth College. Rema Hanna was the coeditor for this article. I thank Oded Galor, Louis Putterman, Jonathan Zinman, Felipe Valencia-Caicedo, Eduardo Montero, Stelios Michalopoulos, Nathan Nunn, Klaus Desmet, Uwe Sunde, Pedro Dal Bó, Rob Blair, Sascha Becker, Roland Bénabou, and Raquel Fernandez for valuable comments and suggestions. I thank participants in the NBER SI on Income Distribution & Macroeconomics, Advances with Field Experiments Conference at the University of Chicago, ASREC Conference at Boston, Deep Determinants in Comparative Development Conference at Brown, NEUDC at Northwestern University, Economics Seminar at Dartmouth College, Seminario CEDE at Universidad de Los Andes, Economics Seminar at Universidad del Rosario, Economics Seminar at Pontificia Universidad Javeriana, and the Macro Seminar and Applied Micro Lunch at Brown University. I also want to thank Juan Uribe, Carolina Lopez, and Patrick Mayero. This project was viable due to the funding provided by the Tinker Foundation, the Andrew Mellon Foundation, and both the Center for Latin American and Caribbean Studies and the Growth Lab at Brown University. Brian Martinez, Rodolfo Mendez, and Daniel Corredor provided outstanding research assistance. The design for the baseline results was preregistered in EGAP under the ID 20180724AB, while the design to assess mechanisms was registered in EGAP under the ID 20190527AA (both designs are currently hosted by the Center for Open Science's OSF Registries). IRB approvals were obtained at Brown University. All mistakes are my own.

[†]Go to <https://doi.org/10.1257/aer.20201332> to visit the article page for additional materials and author disclosure statement.

This paper studies the intergenerational consequences of social exclusion on attitudes and behavior, leveraging a policy that formalized the exclusion of people suffering from leprosy in Colombia. Between 1871 and 1961, the Health Ministry and the Central Hygiene Board in Colombia imposed the isolation of those suffering from leprosy to a colony called Agua de Dios, based on the erroneous belief that the disease was highly contagious and deadly. Physicians appointed by the Central Hygiene Board actively identified and forcefully secluded all sufferers of the disease in the country regardless of their social and economic status. The mistreatment of leprosy sufferers at the hands of doctors continued inside the colony, where doctors experimented with new treatments on sufferers and impinged upon their daily lives. Since the end of the policy of isolation in 1961, leprosy and its associated social exclusion have vanished from the region as the disease and the stigma associated with it have not afflicted new generations.

Despite the notoriety of the policy dealing with leprosy in Colombia, countries across the globe followed similar guidelines when dealing with those suffering from leprosy, enforcing policies that isolated and further stigmatized them. These similarities notwithstanding, the end of the policy was different in Colombia in important ways. As opposed to other countries, the vanishment of leprosy in Colombia did not lead to the relocation of former patients or the disappearance of the community that had been confined. The former colony of Agua de Dios acquired full status as a municipality, leading to the establishment of a community that shared a history of exclusion despite the gradual disappearance of the affliction and the stigma associated with it. New generations in this community did not suffer from the disease or any trait that could be associated with it, which allowed them to fully integrate with descendants of the nonexcluded who lived in surrounding areas. Close study of this historical episode's legacy thus sheds light on the implications of ancestors' social exclusion for the behavior of their descendants, even when the latter are no longer subjected to the mistreatment and segregation endured by the former. Moreover, it does so for a specific type of social exclusion that has been largely overlooked in economics despite its historical and geographical prevalence.

Between 2018 and 2021, I collected experimental and survey data in and around Colombia's former leper colony to examine the differences in social preferences between descendants of excluded locals and descendants of the nonexcluded. Using dictator games and survey data, I find that those with an excluded ancestor display a significantly higher prosociality (altruism and trust) toward in-groups. Moreover, they extend such prosocial attitudes to out-groups who have experienced hardships or circumstances similar to theirs. This finding suggests that for these descendants, the boundaries of group membership may partially expand to incorporate those who share the burden of suffering or trauma, broadly defined. This legacy of prosociality, however, does not extend to all outsiders. Indeed, survey evidence coupled with a novel incentivized measure of trust in medicine suggest that descendants of the excluded mistrust doctors. The direct role of physicians in excluding those suffering from leprosy elucidates the adverse legacy of mistrust toward outsiders who were exclusionary in the past.

I consider several hypotheses that could affect the internal validity of my results and find little support for these explanations. One concern is that the selection of attributes among those who were forcefully sent to the colony explains the patterns

described above. This possibility is mitigated by the fact that leprosy has a long incubation period, coupled with the historical context—a rural economy before the epidemiological transition and before the dissemination of basic sanitary protocols. Combined, these factors make it unlikely that the isolation policy prevented the disease from reaching all segments of society, which reduces the scope for the disease to afflict individuals based on income or education. Indeed, data collected by the colony's chief doctor show that patients from urban and wealthy regions were not over- or underrepresented in the colony.

A second concern is that even if ancestors were not selected, descendants of the excluded may have experienced different opportunities and environments than descendants of the nonexcluded, making it difficult to ascertain that the behavioral patterns are stemming from ancestral exclusion. Census data collected shortly after the end of the isolation policy suggest that this was unlikely the case because secluded locals exhibited levels of education, income, and demographic characteristics that were comparable to those of nonexcluded locals. Hence, participants in my study (who were born after the policy of isolation) grew up facing similar conditions and opportunities regardless of whether their ancestors were excluded or not.

A third concern stems from selective in- or out-migration: those with different prosocial attitudes could be precisely the ones selecting to migrate to, or from, the former colony's region. Reassuringly, out-migrants from the former leper colony exhibit observable attributes similar to those of out-migrants from the surrounding area, and their reported reasons for migrating are both nearly identical and orthogonal to the behavioral outcomes assessed in this paper. Moreover, the average behavior of descendants of the nonexcluded in the study closely resembles the average behavior of respondents in different regions of Colombia, whereas descendants of the excluded (brought there by force) exhibit distinct behaviors compared to Colombians from these other regions. All in all, these findings suggest that selective migration is also an unlikely explanation for the patterns.

I next explore mechanisms using several complementary approaches. Given the scope for narratives to make sense of group experiences and influence actions (Bénabou, Falk, and Tirole 2020; Michalopoulos and Xue 2021), coupled with the ability of active collective memory to shape attitudes and behavior (Dessí 2008; Ochsner and Roesel 2019), I first examine the behavioral effects of historical narratives about the leper colony. Specifically, I exploit a randomization exercise in which participants in the study's region were exposed to different historical narratives about the social exclusion of leprosy sufferers. These treatments were designed with the help of a local oral historian and with the explicit goal of highlighting narratives likely persuasive to participants with an excluded ancestor. I find that descendants of the excluded who were exposed to a locally salient narrative highlighting the banishment and isolation of their ancestors display significantly higher in-group altruism. On the other hand, those exposed to a locally salient narrative highlighting the historical medical misperceptions about leprosy display lower trust in medicine.

To validate the randomization exercise and to further investigate the content of local historical narratives, I surveyed a different set of participants two years after the intervention. Descendants of the excluded who participated in this survey were more likely to share historical narratives that mentioned the stigma and mistreatment endured by leprosy sufferers. They were also more likely to mention that the

historical actions of doctors were grounded on misinformation, which coincides with the interventions I had implemented two years before. I also asked participants in this subsequent round of fieldwork about the sources from which they learned the historical information they had just shared. Nearly 60 percent of those with an excluded ancestor reported that their direct ancestors (parents and grandparents) were a relevant source. Conversely, sources that can be categorized as institutional, horizontal, or oblique (e.g., schools, friends, or neighbors) were not frequently mentioned by descendants of the excluded. The sources of historical narratives for those who did not descend from the excluded are notably different: oblique and horizontal sources figure more prominently, while only 20 percent reported direct ancestors as relevant sources. This suggests that the content of historical narratives from excluded ancestors help shape the behaviors of their descendants, as they are more persuasive at raising the saliency of exclusion.

This paper contributes to several strands of literature. First, it adds to the literature that examines the consequences of historical episodes, particularly traumatic ones, on beliefs and attitudes (Nunn and Wantchekon 2011; Lowes and Montero 2021a; Nunn 2021; Williams 2022). The paper presents evidence of the lasting consequences of social exclusion on the strengthening of parochial prosociality, the formation of durable in- and out-group distinctions, and the emergence of persistent mistrust toward those who were exclusionary in the past. In doing so, it also contributes to the burgeoning literature that employs lab-in-the-field approaches to study the legacy of historical phenomena (Lowes et al. 2017; Lowes 2018, 2020; Karaja and Rubin 2022). This study implements such an approach to examine the exclusion that has occurred in leper colonies—an understudied geographical pattern of exclusion that has occurred since antiquity in various regions of the world.

The paper's results are also directly related to findings from recent papers that study the lasting adverse effects of nonconsensual medical interventions, which have been documented both in developed and developing regions (Alsan and Wanamaker 2018; Lowes and Montero 2021b; Martinez-Bravo and Stegmann 2022). Indeed, the paper presents another instance of an adverse legacy of mistrust in medicine that stems from medical malpractice. Moreover, it contributes with a novel revealed-preference measure of trust in medicine that can be adapted and implemented in other contexts: the take-up of a medicine recommended by doctors, which participants could redeem for free if they simply visited the local pharmacy.

The paper also contributes to the literature on the behavioral consequences of social exclusion. Although understudied in economics, these implications have been investigated by other disciplines. In particular, research in social psychology has found that socially excluded people tend to avoid interacting with those who excluded them while being more likely to display empathy and altruism, particularly toward underprivileged groups (Maner et al. 2007; Vollhardt and Staub 2011; Greenberg et al. 2018). This study presents new evidence that such behavioral patterns may persist across generations, even when the exclusion was only endured by ancestors.

This paper's results also speak to the literature that investigates the roots and consequences of differences in prosociality toward in- and out-groups (Petersen 2001; Wood 2003; Bowles 2006; Choi and Bowles 2007; Rohner, Thoenig, and Zilibotti

2018; Lowes, Rosignol, and Nunn 2022; Enke, Rodriquez-Padilla, and Zimmermann 2022a, b). It shows that the historical experiences of social exclusion may carry a complex legacy on prosocial attitudes toward different recipients: descendants of the excluded display higher in-group prosociality, extend such prosociality toward outsiders who have undergone similar experiences, and display lower trust toward those perceived as members of the exclusionary group. Historical social exclusion may thus be one of the factors that explains why some demographics behave differently toward in-groups than out-groups.

Finally, the paper also furthers the investigation into the transmission and behavioral effects of specific cultural practices, such as narratives (Bisin and Verdier 2001; Bénabou, Falk, and Tirole 2020; Michalopoulos and Xue 2021). Relatedly, it contributes to the literature that examines how collective memory is activated to affect views and beliefs (Dessí 2008; Ochsner and Roesel 2019; Bordalo, Gennaioli, and Shleifer 2020). The paper leverages open-ended responses to examine the content of historical narratives about experiences endured by ancestors and combines it with an experimental approach to examine the extent to which these narratives affect behavior. It thus exploits a novel empirical approach that advances our understanding of narratives as cultural vehicles that contribute to the persistence of historical episodes.

The remaining sections are structured as follows. Section I provides the historical background on the stigma leprosy sufferers have been subject to, along with the specificities of the leper colony of Agua de Dios. Section II describes the empirical approach, while Section III describes the recruitment process and the variables collected. Section IV presents the main results, Section V examines historical narratives as an intervening mechanism, and Section VI concludes.

I. Leprosy and Its History

Leprosy is a long-term bacterial infection believed to be transmitted mainly through nasal droplets. Formally known as Hansen's disease, leprosy's main symptoms are neuropathic pain, sensory loss in the extremities, skin lesions, and cartilage deformation (World Health Organization 2022). Although close contact with an infected individual is the most critical risk factor, it is not a highly infectious disease (Schreuder, Noto, and Richardus 2016). Exposure to the bacteria is a necessary but insufficient condition for infection; evidence suggests that there is a genetic predisposition to contract and physically manifest leprosy (Abel et al. 1998; Mira et al. 2003; Misch et al. 2008). Hence, although a genetic component may affect the proclivity to manifest leprosy, the disease is not genetically transmitted.

Furthermore, the disease is not sexually transmitted, transmitted from mothers to their unborn children, or deadly (Centers for Disease Control and Prevention 2017a). It initially presents no manifestation and can remain dormant for more than two decades (World Health Organization 2022), and in most cases the immune system can eliminate the infection before any physical manifestation occurs. The disease is currently treated with multidrug therapy, which combines different types of antibiotics such as dapsone, rifampicin, and clofazimine (Centers for Disease Control and Prevention 2017b). Treatment with dapson dates back to the 1940s; before then, leprosy was treated either informally or with treatments that had limited

success such as chaulmoogra oil (Bennet, Parker, and Robson 2008). If untreated and if the immune system does not eliminate the bacteria, leprosy may reach an advanced stage characterized by nerve damage, ulcers, and lack of sensation that may further the prevalence and severity of skin lesions. This advanced, visually manifested stage of the disease has been historically subject to social stigma and exclusion around the world.

Sufferers of leprosy have been historically stigmatized and socially excluded. The mistreatment of these individuals is nonracial and has been prevalent across geographies and cultural traditions: Judeo-Christianity, Buddhism, and Hinduism envision the disease as a manifestation of sin, bad karma, or inherent pollution (Buckingham 2002). The stigma against sufferers is thus not specific to Catholicism, the Western world, or to a specific epoch. Even though this mistreatment was globally prevalent, the first institutions destined to segregate these individuals occurred in medieval Europe, with sufferers increasingly forced to live in specific houses. These housing units, known as leprosaria, appeared throughout Europe during the Middle Ages (Miler and Nesbit 2014). Negative perceptions of, and aversion to, those with leprosy intensified with the Black Plague, resulting in a public administration that furthered their social exclusion through stricter physical segregation.

With the advent of colonial domination, the administration of leprosy through isolation was quickly adopted around the world as this institutional design formalized an enduring, globally prevalent cultural stigma. The increasing numbers of leprosy sufferers required larger spaces in which to house them, causing leprosaria to gradually transform into colonies that housed hundreds if not thousands of people with the disease. This transition coincided with the advent of modern medicine, which increasingly played a central role in dealing with leprosy, identifying its cases, and managing its sufferers' symptoms. Moreover, physicians legitimized and endorsed the existence of leper colonies, conceptualizing the disease as an inherent risk to the health and even the survivability of healthy subjects. Indeed, the European Medical Association and subsequently the First Congress on Leprosy in 1897 declared the disease to be dangerous and highly contagious, promoting leper colonies as a vital component of the strategy to eradicate the disease (Obregon 2004).

The proliferation of leper colonies after the second half of the nineteenth century followed a medical model aimed at containing leprosy as a disease. Based on the inaccurate medical consensus that leprosy was highly contagious and potentially deadly, national medical entities and the physicians that represented them were far more concerned with the perceived public health benefits of isolating the ill and finding a cure for leprosy than with the individual well-being of patients (Obregon 2004; Platarueda Vanegas 2008). The active involvement of physicians, forceful displacement of patients, strict segregation, and experimentation with the ill were thus common features of leper colonies on a global scale (Kikuchi 1997; Buckingham 2002; Tayman 2007; Arcilla 2009; Manton 2011).

Online Appendix A.1 maps the countries that followed the policy of creating leper colonies with these features, showing the global prevalence of this phenomenon. In particular, state medical entities across the Americas played a prominent role in identifying the ill, strictly segregating them to avoid their interactions with other subjects, and interfering with their lives—or outright experimenting with

them once in isolation. This was the case with leper colonies in the United States (Mizell-Nelson 2003), Uruguay (Imprenta Nacional República Oriental de Uruguay 1938), Peru (Cueto 2004), Venezuela (Romero Salazar 2001), Argentina (Senado y Cámara de la Nación Argentina 1928), and Brazil (Modelli 2016). Colombia, the case studied in this paper, conformed to the principles that governed leper colonies globally and in the Americas.

A. The Leper Colony of Agua de Dios

In 1871, the Colombian government officialized the foundation of the leper colony of Agua de Dios. Although the colony was initially meant to house all cases of leprosy from Colombia's central region, by the 1940s it was receiving cases identified throughout the country. The Central Hygiene Board—an institution run by doctors who followed the guidelines of the European Medical Association, the First Congress on Leprosy, and the Medical Academy of Bogotá—was the entity responsible for establishing how to identify and handle leprosy sufferers. A series of rules and procedures were set at the national level to ensure that all leprosy sufferers in Colombia were effectively secluded. All citizens legally had to report anyone who suffered from any deformation or physical manifestation akin to leprosy (Lazaretto National Direction 1927). The town/city in which such a report occurred had to appoint a physician who could examine both the suspect and close contacts. Only those who tested positive were forcefully sent to the leper colony, while no one who tested negative could enter (Gutiérrez Pérez 1925; Lazaretto National Direction 1927). This test reinforced the trauma experienced by those who had leprosy as most sufferers had to endure family separations before being interned.

Secluded individuals lost their citizenship as soon as they entered the colony, with doctors replacing their identification cards with documentation that certified them as perpetual patients. All subjects were welcomed by doctors, who assessed and monitored all individuals (Oyaga 2013). Fences surrounding the colony, combined with military supervision from outside, enforced the isolation (Lazaretto National Direction 1927). The medical administration of the ill continued within the colony as doctors imposed strict protocols on patients, including prohibiting marriages and marital relations based on the misperception that mothers would transmit the disease to their offspring (Martínez Morales 2001).

Further, with the lack of any cure for the disease or treatment to reverse the symptoms, physicians occasionally experimented alternative modalities on patients (Oyaga 2013). From patients' perspectives, physicians thus became visible actors responsible for their exclusion and mistreatment. No other public official operated within the colony besides government-appointed physicians, which furthered the patients' perception that doctors were responsible for their isolation. Aside from physicians, only members of religious orders assisting the ill and authorized merchants were allowed temporarily in the colony (Ramos Arenas 2008).

Data collected by the chief doctor of the colony in 1920 reveal that 90 percent of the isolated population came from rural areas, a figure close to the overall share of Colombia's rural population at the time. The overwhelming majority of the secluded population was composed of former agricultural peasants, which was also the predominant occupation at the time both in Colombia and in the

region where the colony was located (Gutierrez Perez 1925). Patients came from across the central region of the country, which was homogeneous along religious (Catholicism), ethnic (mestizos), and linguistic (Spanish) lines, meaning that the colony's population was also homogeneous in these dimensions (Aparicio 1919; Gutierrez Perez 1925).

Online Appendix A.2 maps the municipalities of origin of all individuals isolated in Agua de Dios in 1920, as recorded by the colony's chief doctor at the time. The number of patients from each municipality of origin can be used to measure the incidence of leprosy in the municipalities belonging to the central region of Colombia, where recorded case of leprosy would have been sent to the colony. This information allows for examining whether municipalities where leprosy was severe were different than other municipalities along observable dimensions.

Online Appendix A.3 shows that the incidence of leprosy at the time was not meaningfully associated with economic development (as proxied by population density), gender composition, or conflict, some of the few historical variables that are available for municipalities across Colombia's central region. Moreover, the fact that the incidence is not significantly associated with population density also suggests that the disease was not predicted by lack of historical sanitation, which was historically concentrated in places with high population densities (Lleras Camargo 1997; Rodríguez Leuro 2014). These results give credence to the claim that, in the context of a rurally poor economy before the epidemiological transition (such as in Colombia), the disease did not disproportionately affect wealthier or socioeconomically different municipalities.

The trauma and strict segregation experienced by patients at the hands of doctors contrasted with the openness and generosity with which patients received the incoming population despite being strangers to them. Those already secluded often shared their homes with those who did not have anywhere to stay, and others generously helped the vulnerable by assisting in building houses and essential infrastructure (Ramos Arenas 2008; Oyaga 2013; Alcaldía Municipal de Agua de Dios 2017). All in all, the isolated population developed altruism and solidarity to resist and counter its social exclusion (Platarueda Vanegas 2008, 2019). Hence, the shared experience of social exclusion plausibly triggered the development of altruistic behaviors capable of mitigating the adverse experience shared by all members of the group.

Although the Colombian state legally abolished the policy of isolation in 1961, some of the colony's restrictions began to ease starting in 1950. Former patients were, starting that decade, gradually allowed to form families, have children, and reunite with relatives, some of whom migrated to the former colony (Platarueda Vanegas 2019). Given that new generations did not suffer from the disease, descendants of the excluded started to populate surrounding areas, integrate with people living in them, and vice versa. To date, leprosy has nearly vanished from Colombia and from Agua de Dios, and these descendants were born and raised facing similar conditions and opportunities as their local peers who were not descendants of the excluded. Despite no longer being subjected to stigma or exclusion, descendants of the excluded grew up keenly aware of the history of suffering and apartheid endured by their ancestors, partly due to the relevance of oral histories in such a context (Botero-Jaramillo, Mora-Blanco, and Quesada-Jimenez 2017).

II. Empirical Approach

I focus on the descendants of those who were excluded due to leprosy to assess social exclusion's long-term consequences on different dimensions of prosociality. The history of Agua de Dios suggests that the mistreatment experienced by those with the disease may have fostered local prosociality to counteract such a trauma. Additionally, such traumatic experiences may have also led to a greater concern for the well-being and suffering of others (Vollhardt and Staub 2011; Greenberg et al. 2018). On the other hand, the responsibility of various actors in excluding leprosy sufferers may have left a durable mark on mistrust toward some outsiders, particularly doctors. This complex legacy of social exclusion underscores the importance of measuring attitudes toward locals and different outsiders, particularly physicians and other groups that have suffered.

Between 2018 and 2021, I collected survey and experimental data in the historical region of the most populous (and eventually the only functioning) leper colony in Colombia, leveraging the fact that most of the leprosy sufferers remained in this region once their isolation ended. Those suffering from visible leprosy could not easily conceal the physical attributes that triggered their rejection by out-groups. However, their descendants did not suffer the disease and held the same ethnic, linguistic, and religious identities as those in the surrounding area, allowing them to integrate with locals in the region while avoiding the isolation and stigma that their ancestors endured. Collecting information from this setting allows me to compare individuals in the same environment with different ancestral experiences, which is methodologically akin to an epidemiological approach to studying culture (Fernandez 2007). To the extent that these individuals were born in the same environment, facing similar socioeconomic opportunities, coupled with the fact that the exclusion endured by leprosy sufferers is several decades in the past, comparing the responses and choices of these subjects provides a window into the intergenerational consequences of social exclusion on behavior. Importantly, this assessment affords an opportunity to examine whether a legacy may persist despite the fact that descendants have not experienced the social exclusion endured by their ancestors.

The study's setting is the region of the largest leper colony in Colombia, Agua de Dios. Even though the country initially had three leper colonies, the other two were gradually closed during the 1940s and the state relocated most of their patients to Agua de Dios. Figure 1 maps the municipalities in which I collected experimental and survey information. I focus on these municipalities because a vast majority of those historically linked to the former leper colony still live in them. According to the 1973 census, 88 percent of those who reported being born in or being originally from Agua de Dios lived in the municipalities where I conducted fieldwork (Agua de Dios, Ricaurte, and Tocaima). Similarly, the 2005 census reveals that 86 percent of those who reported having their origins in Agua de Dios lived in these municipalities, shaded in the figure.

Figure 1 also shows another municipality that adjoins the former leper colony, in white (Nilo). Although geographically close, only 0.2 percent of those who reported being originally from or born in the former leper colony lived in this fourth municipality in 1973. Similarly, by 2005, only 0.5 percent of those who reported being from Agua de Dios lived in this fourth municipality. These data underscore the

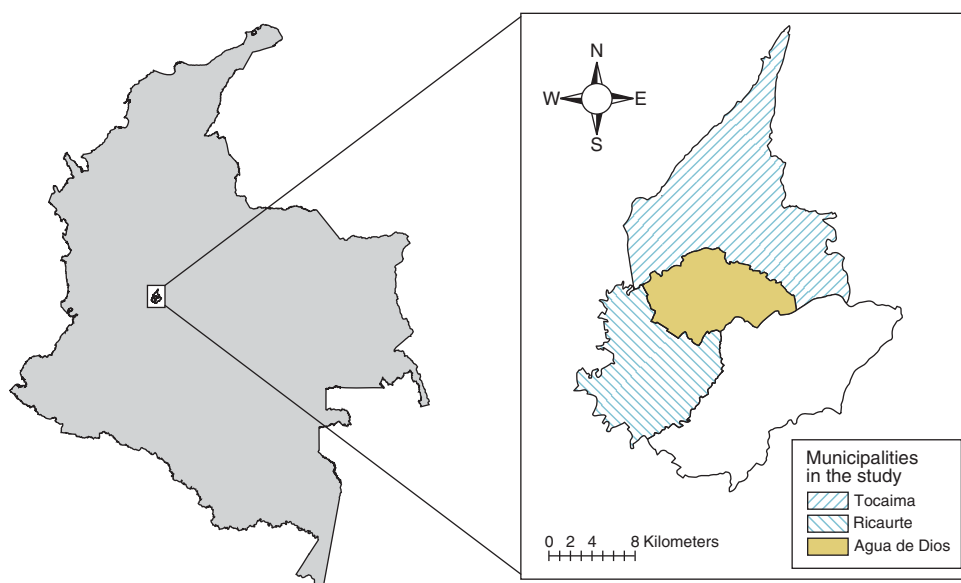


FIGURE 1. LOCATION OF AGUA DE DIOS AND ITS SURROUNDING MUNICIPALITIES

Notes: The map depicts the location and the spatial arrangement of the former leper colony, Agua de Dios (with an area of 86 square kilometers), and its contiguous municipalities where data was collected for this study (Tocaima and Ricaurte with areas of 246 and 128 square kilometers, respectively). Nilo, in white, is the third adjoining municipality, where no data were collected.

relevance of conducting fieldwork in the three municipalities shaded in the figure instead of the fourth one, where recruiting participants with an ancestral link to the leper colony would have been improbable.

Moreover, as shown in online Appendix A.5, the adjoining municipality where I did not collect data was historically poorer than the region of study, with lower rates of literacy, lower educational attainment, and a higher incidence of poverty. Hence, its current inhabitants are descendants of people who experienced different socioeconomic circumstances than those faced by the socially excluded and their descendants. In what follows, I will not be referring to this omitted adjoining municipality whenever I refer to municipalities that are contiguous to (or are surrounding) the former leper colony.

Those who lived in isolation experienced socioeconomic conditions similar to those experienced by the nonexcluded who were native to the region where I collected data. The chief doctor of the leper colony recorded in 1920 the municipality of origin of every patient in the colony. This information can be combined with the limited data available in the 1918 census (female and male counts at the municipality level) to examine whether the population of secluded people in the colony were historically similar to the population in surrounding municipalities. As shown in online Appendix A.6, the male share and population density in the leper colony were very close to those in the municipalities surrounding it. Moreover, the male share and population density predicted by the municipalities of origin (as predicted by the weighted average of these variables, weighted by the share of isolated people from

each municipality of origin) are nearly identical to the ones observed in both Agua de Dios and its surrounding municipalities.

Further, as discussed in the previous section, the chief doctor also noted that the overwhelming majority of the colony's population was composed of agricultural peasants, which was also the predominant occupation at the time in the municipalities surrounding the colony. These historical sources from the epoch of isolation thus suggest that the population secluded in the leper colony came from municipalities similar, on average, to the municipalities surrounding the colony where I collected data. Moreover, they suggest that these populations were similar to one another along observable attributes—a pattern that will also hold in subsequent years for which individual-level data on a wider set of characteristics are available.

Table 1 presents data from the first census for which there are microdata from individuals who are local to the region of study. Even though the census lacks information on whether an individual was excluded because of leprosy or not, it does contain information about an individual's municipality of birth and the municipality where they are from (regardless of the municipality of residence at the time of the census). As an approximation to examining differences between the population that was historically excluded and locals who were historically not excluded, the table compares those who reported being born in or being from the former leper colony with those who report being born in or being from the municipalities contiguous to the former colony. The table shows that those who likely experienced the exclusion and locals who likely did not were similar along an array of observables. Importantly, those who likely endured the historical exclusion exhibit comparable educational attainment, wealth, and child mortality, suggesting that inhabitants in the colony were largely exposed to similar socioeconomic conditions as those in the surrounding municipalities of Ricaurte and Tocaima.

In online Appendix A.7, I restrict the comparison shown in Table 1 to those born in Agua de Dios or in its contiguous municipalities between 1950 and 1961. Given that having children was strictly forbidden for the excluded before 1950, coupled with the fact that full legal rights were not granted to the former leper colony until 1961, there is a higher certainty that those born in the colony during this time frame had to live under the policy of exclusion as opposed to those born in the contiguous municipalities during the same time frame. Reassuringly, the results presented in the table suggest that, even when focusing on a restricted sample for which there is higher certainty of the excluded status, the socioeconomic conditions experienced by excluded and nonexcluded people in the region of study were similar.

Table 1 also presents detailed data from the region about housing units in 1973. Ideally, one would compare the characteristics of housing units owned by excluded locals with the characteristics of housing units owned by the nonexcluded. Although these data do not exist, one can compare the characteristics of housing units inside the former leper colony with those in the surrounding municipalities. To the extent that the units were built before the end of the isolation policy, these comparisons offer further evidence of the similar living standards held by the excluded vis-à-vis that of the nonexcluded. Notably, the characteristics of housing units built in the colony are comparable to those built in the surrounding area, reinforcing the claim that those who lived in isolation were not subjected to systematically different poverty rates.

TABLE 1—CHARACTERISTICS OF THOSE LIKELY EXCLUDED IN THE LEPER COLONY COMPARED TO NONEXCLUDED LOCALS

	Colony = 1 mean (1)	Colony = 0 mean (2)	OLS coefficient on <i>Colony</i> (3)	Standard error (4)	Observations (5)
<i>Panel A. Sociodemographic characteristics: population census, 1973</i>					
Female	0.511	0.510	0.001	(0.016)	51,144
Age	27.639	27.411	0.227	(2.018)	51,144
Marital status	0.244	0.281	−0.037	(0.044)	51,144
Primary	0.760	0.751	0.009	(0.047)	46,045
Secondary	0.205	0.171	0.035	(0.046)	46,045
Labor force participation	0.286	0.371	−0.086	(0.051)	50,226
Children	3.226	4.128	−0.902	(0.371)	14,155
Deceased children	0.328	0.327	0.001	(0.043)	11,097
Literacy	0.786	0.760	0.027	(0.051)	46,046
Income	1,045.853	1,372.335	−326.483	(273.206)	15,036
<i>Panel B. Housing characteristics: housing census, 1973</i>					
Has no ground slab	0.498	0.453	0.045	(0.045)	7,445
Walls made out of bricks	0.396	0.414	−0.018	(0.112)	7,445
Roof with concrete slabs	0.054	0.069	−0.015	(0.030)	7,445
Has potable water	0.486	0.398	0.088	(0.105)	7,445
Has sewerage	0.613	0.336	0.277	(0.173)	7,445
Has a kitchen	0.754	0.788	−0.034	(0.050)	7,445
Has a toilet	0.336	0.278	0.058	(0.116)	7,445
Has electricity	0.785	0.464	0.321	(0.186)	7,445
Own house	0.431	0.420	0.011	(0.024)	7,445
Credit-funded	0.031	0.040	−0.008	(0.012)	7,445

Notes: The table examines the conditions faced by those who were likely excluded in the colony and those faced by locals who were not excluded (those originally from the municipalities of Tocaima and Ricaurte). Panel A examines the population census of 1973 to compare the sociodemographic characteristics of individuals who report being from or being born in Agua de Dios (the former leper colony) and those of individuals who report being from or being born in the surrounding area. Panel B examines the housing census of 1973 to compare the characteristics of housing units built in the colony and those built in the contiguous region. Each row in column 3 comes from a regression of the dependent variable in the row heading on an indicator that equals one for individuals (housing units) that were excluded in the colony (built in the colony), while column 4 presents the corresponding standard errors of such coefficients. Standard errors in panel A are clustered at the municipality of residence in 1973. Standard errors in panel B are clustered at the neighborhood level. Column 5 presents the observations of each regression. Observations in panel A vary due to a different number of responses collected for different variables in the population census of 1973. See online Appendix A.9 for further details of the outcomes described in the row headings. OLS = ordinary least squares.

Potential differences in other dimensions are not relevant in the context of the study. Differences in climatic conditions endured by ancestors are improbable due to the proximity of the municipalities that I examine. Indeed, online Appendix A.8 displays a climatic dissimilarity index that I constructed for each municipality in Colombia, which provides a summary index of the climatic differences between these municipalities and Agua de Dios. Reassuringly, the figure shows that the municipalities contiguous to Agua de Dios exhibit the smallest climatic dissimilarity with respect to the former leper colony. This similarity is also present in other relevant dimensions: linguistic, ethnic, and religious affiliations of ancestors are not

a concern in the context of the study. The lack of differences in these dimensions is explained by the fact that those sequestered in the colony came from the country's central region, which was historically inhabited by mestizos, Spanish speakers, and Catholics. Given that municipalities surrounding the colony were also located in this region, excluded and nonexcluded locals were homogeneous along these lines.

In sum, ancestors of individuals in the region of study lived in similar environments and held similar demographic and economic attributes, a finding that reinforces the historical resemblance of these municipalities discussed in Table 1 and online Appendix A.6. These results suggest that study participants grew up in comparable socioeconomic environments and faced similar conditions and opportunities, regardless of whether their ancestors were excluded due to leprosy or not. Moreover, descendants of the excluded avoided the stigma and rejection experienced by their ancestors given that they did not suffer from leprosy and held the same ethnic, religious, and linguistic affiliation as other locals.

III. Data Collection

I collected survey and experimental data in three rounds of fieldwork, surveying a different cross section of participants in the region of study in 2018, 2019, and 2021. Every time I collected data, there was no expectation (by the research team, the people in the area, or even me) that we would come back to the region at a later date or that the research team would recontact participants. In all rounds, I recruited participants following a method akin to a random walk: with the help of local leaders, all the neighborhoods within each municipality were identified, along with central locations within each of those neighborhoods. Starting from these nodes, the enumerators selected arbitrarily one path and invited one member per household they approached.

Each round of fieldwork served a different purpose: I collected most of the outcomes for the baseline results in the first round, whereas outcomes from the second and third rounds were mostly meant to assess mechanisms. Participants in rounds one and three of fieldwork provided responses without prior interventions, whereas participants in round two were randomly exposed to different historical narratives before providing their responses (Section V and online Appendix A.34 provide further details on this randomization exercise). Moreover, participants in round three were asked to share what they knew about doctors' historical role in the leper colony as well as the sources that informed these historical views (both questions were open-ended, and participants could share whatever they chose to).¹

The outcomes from two rounds of dictator games provided the experimental measures of prosociality (Kahneman, Knetsch, and Thaler 1986; Forsythe, Horowitz, and Savin 1994). The game entails a one-shot interaction, where subjects choose how to divide a given sum of money between themselves and a second randomly chosen, anonymous individual. Study participants played two rounds of the games: one

¹Differences in the number of observations across rounds did not reflect locals' decisions or engagement but rather the conditions that the research project faced in different points in time (e.g., funding available in different stages of the project, or the number of observations deemed feasible by a survey company due to the pandemic). Online Appendix A.34 describes the recruitment method and the details of the protocol that I implemented in each round of fieldwork.

in which the anonymous recipient was someone from the same municipality and another in which the recipient was from a distant municipality. I implemented slightly different versions of this game in the first and second rounds of fieldwork.

In the first round of fieldwork, which I assess in the baseline results, participants received physical money and were given discretion in deciding how much of that money they wanted to keep for themselves and how much they wanted to give to the anonymous receiver. On that occasion, they were given Col\$16,000 per round (8 bills of Col\$2,000, which amounted to approximately US\$6 per round at the time). Hence, participants in the first round of fieldwork got to allocate at most Col\$16,000 to in-groups and at most Col\$16,000 to out-groups. In this round of fieldwork, the out-group recipient was someone from Apulo, a municipality explicitly selected with the goal of providing a nearby out-group that was similar in socioeconomic characteristics to the region of study.

In the second round of fieldwork, participants automatically entered a lottery for Col\$800,000 (approximately US\$250 at the time). They were then asked how they would split that prize between an anonymous receiver from the same municipality, an anonymous receiver from a distant municipality, and themselves. On this occasion, participants were interviewed at their doors, reported the amounts they wanted for each party, and were warned that their choices would be enforced if they won the lottery.² Another important difference in the second set of dictator games was the out-group: these recipients were from Pandi, a municipality selected with the explicit goal of providing an out-group that was also similar to the region of study but more distant than out-groups from the first set of dictator games.³ Online Appendix A.4 maps the out-groups of both rounds of fieldwork, Apulo and Pandi, and online Appendix A.5 shows that these municipalities were historically similar to the region of study across an array of economic and sociodemographic characteristics.

In each round of fieldwork, I asked participants whether any of their ancestors lived in the colony during the epoch of isolation. To avoid biasing their responses, ancestral status was the last piece of information collected in all rounds, so the balance of participants' demographic characteristics across ancestral lines was not targeted. Table 2 compares participants who have an excluded ancestor with those who do not. These participants are similar across most observable dimensions, although the differences in secondary attainment across rounds highlight the importance of discussing results with and without controlling for these observable factors. Importantly, however, these differences do not reflect a historical imbalance in educational attainment, as seen in Table 1. Moreover, Section IV discusses robustness checks that examine (and reduce the scope for) education as a mediating channel.

The results that follow are based on a self-reported measure of ancestry.⁴ Even though this opens the possibility for selective reporting, I undertook additional steps in the second round of fieldwork to increase the odds of accurate responses. On that

²In the first and second set of dictator games, there was no deception and the allocations chosen by participants were honored. See online Appendix A.34 for a discussion of these and other details of each of the preregistered protocols.

³The distance between the former leper colony and Apulo is 23 kilometers; it takes approximately 5 walking hours to cover this distance. Pandi is substantially farther from it than the out-group from the first round of fieldwork: 63 kilometers and 13 walking hours away from the former leper colony.

⁴Online Appendix A.9 describes and defines the variables employed in the analysis, while online Appendix A.10 presents summary statistics of all the variables.

TABLE 2—DEMOGRAPHIC CHARACTERISTICS OF STUDY PARTICIPANTS BY EXCLUDED-ANCESTRY STATUS

	Ancestry = 1 mean (1)	Ancestry = 0 mean (2)	OLS coefficient on <i>ancestry</i> (3)	Standard error (4)	Observations (5)
<i>Panel A. First round of fieldwork, 2018</i>					
Female	0.585	0.599	−0.014	(0.061)	265
Age	43.085	43.252	−0.167	(2.074)	265
Primary	0.856	0.816	0.040	(0.046)	265
Secondary	0.627	0.497	0.131	(0.061)	265
Marital status	0.492	0.578	−0.087	(0.062)	265
Children	1.720	2.190	−0.470	(0.232)	265
Total	118	147			265
<i>Panel B. Second round of fieldwork, 2019</i>					
Female	0.519	0.497	0.022	(0.053)	360
Age	46.962	46.407	0.555	(1.842)	360
Primary	0.858	0.808	0.050	(0.039)	360
Secondary	0.650	0.537	0.114	(0.052)	360
Marital status	0.514	0.486	0.028	(0.053)	360
Children	1.978	1.994	−0.016	(0.181)	360
Total	183	177			360
<i>Panel C. Third round of fieldwork, 2021</i>					
Female	0.596	0.530	0.065	(0.063)	275
Age	46.340	45.790	0.550	(2.266)	275
Primary	0.968	0.862	0.106	(0.032)	275
Secondary	0.691	0.525	0.167	(0.061)	275
Marital status	0.479	0.442	0.037	(0.064)	275
Children	0.766	0.740	0.026	(0.055)	275
Low income	0.165	0.245	−0.081	(0.053)	248
Total	94	181			275

Notes: The table examines the demographic characteristics of study participants in the first (panel A), second (panel B), and third (panel C) rounds of fieldwork. A different pool of participants was recruited in each round. Columns 1 and 2 show the mean value of each variable described in the row headings for participants with and without an excluded ancestor, respectively. Each row in column 3 comes from a regression of the dependent variable in the row heading on an indicator that equals one for participants who are descendants of those excluded in the colony, while column 4 presents the corresponding robust standard errors of such coefficients. Column 5 shows the number of observations of each regression. The observations vary across panels due to the different number of participants recruited or surveyed in each round.

occasion, enumerators told participants that receiving the lottery prize was contingent on demonstrating that their reported ancestry was true, thereby ensuring that accuracy was incentivized.

In online Appendix A.11, I compare the observable attributes of participants in the second round who reported having an excluded ancestor (when the odds of accuracy were higher due to the incentives) with those who reported the same in other rounds of fieldwork. Likewise, I examine the same comparison for those who reported not having an excluded ancestor. Notably, the table indicates no meaningful observable differences between those who reported having an excluded ancestor in the second

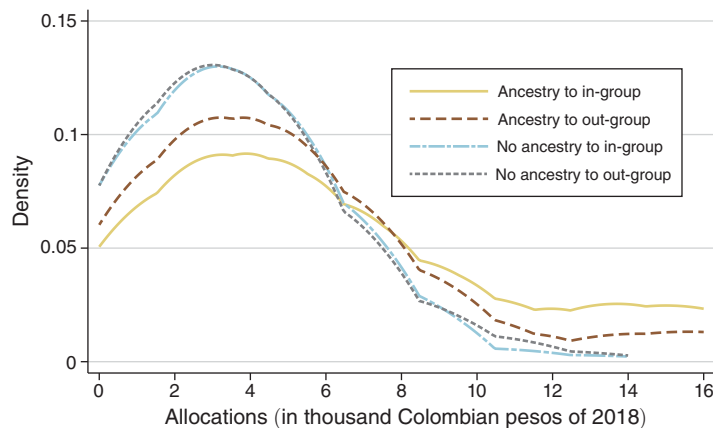


FIGURE 2. DISTRIBUTIONS OF ALLOCATIONS IN DICTATOR GAMES IN THE FIRST ROUND OF FIELDWORK

Notes: The figure displays the distributions of the allocations that participants in the dictator games of the first round of fieldwork gave to in-group and out-group recipients, split by ancestral status. Ancestry refers to descendants of people that lived in the colony during the epoch of isolation. No ancestry refers to descendants of people that did not live in the colony during the epoch of isolation. In-group refers to anonymous receivers from the same municipality as the respondent, whereas out-group refers to anonymous receivers in Apulo, a municipality outside the region of study. Each participant allocated up to Col\$16,000 to the anonymous in-group and up to Col\$16,000 to the anonymous out-group (US\$1 was approximately Col\$2,900 at the time, and Col\$16,000 amounted to half of a daily minimum wage).

round and those who did the same in other rounds; the same pattern holds when comparing those who reported having no excluded ancestor in the second versus other rounds of fieldwork. To the extent that the incentive provided on the second round of fieldwork was effective at improving accurate information about ancestry, this evidence suggests that the scope for selective reporting is limited.

IV. Results

I use a combination of survey and experimental outcomes to examine differences in prosociality across excluded ancestry status. Figure 2 shows the distribution of the allocations that participants gave to in- and out-groups in the dictator games conducted during the first round of fieldwork, split by excluded ancestry status. Notably, the distributions of allocations for descendants of the excluded suggest a systematically higher altruism toward both types of recipients.⁵ Moreover, these distributions suggest that they display an even higher altruism toward in-group members.

To further examine this legacy of prosociality, I estimate the following expression with different dimensions of altruism as outcome variables:

$$(1) \quad Y_i = \alpha + \beta \text{Ancestry}_i + \vartheta' \mathbf{X}_i + \epsilon_i.$$

In expression (1), Y_i is the outcome of interest for individual i , which includes

⁵In online Appendix A.12, I validate participants' behaviors in dictator games and show that self-reported altruism robustly predicts higher in-group and out-group allocations.

TABLE 3—DIFFERENCES IN ALTRUISM BETWEEN STUDY PARTICIPANTS WITH AND WITHOUT AN EXCLUDED ANCESTOR

	Dictator games				Survey measures			
	In-group allocation (1)	Out-group allocation (2)	Total allocation (3)	In-group–out-group allocation (4)	Altruism Venezuelan refugees (5)	Large business owners (6)	Self-reported altruism (7)	ICW index altruism (8)
<i>Panel A. No covariates</i>								
<i>Ancestry</i>	2.680 (0.488)	1.401 (0.429)	4.081 (0.836)	1.278 (0.382)	0.888 (0.283)	0.640 (0.329)	0.589 (0.235)	0.424 (0.072)
Observations	265	265	265	265	265	275	265	265
R^2	0.117	0.043	0.093	0.047	0.035	0.015	0.022	0.119
<i>Panel B. With baseline covariates</i>								
<i>Ancestry</i>	2.583 (0.502)	1.296 (0.445)	3.879 (0.865)	1.287 (0.392)	0.968 (0.294)	0.700 (0.330)	0.690 (0.243)	0.436 (0.075)
Observations	265	265	265	265	265	275	265	265
R^2	0.140	0.066	0.109	0.107	0.082	0.058	0.092	0.155
Mean dep. var.	4.649	4.189	8.838	0.460	7.491	2.898	7.970	0.000
Round 1	✓	✓	✓	✓	✓		✓	✓
Round 2								
Round 3						✓		

Notes: The table documents the legacy of ancestral social exclusion on various measures of altruism. Each panel-column reports results from a single OLS regression of the dependent variable described in the column heading on the variable(s) described in the row heading(s). Panel B shows the results of regressions that control for gender, age, age squared, number of children, marital status, and indicators for primary and secondary education. *Ancestry* is an indicator that equals one for participants who reported that they have an ancestor who lived in the colony during the epoch of isolation. In-group allocation is the amount given by each participant in a dictator game to an anonymous receiver from the same municipality. Out-group allocation is the amount given to an anonymous receiver from Apulo, a municipality outside the region where the dictator games were conducted. Total allocation is the sum of what each participant gave in both rounds of dictator games. In-group–out-group allocation is the difference between what a participant assigned to the in-group and the out-group recipients. Allocations are in thousand Colombian pesos of 2018, and participants allocated 16,000 at most to each type of recipient (US\$1 was approximately Col\$2,900 at the time, and Col\$16,000 amounted to half of a daily minimum wage). Altruism Venezuelan refugees is the reported support for policies to assist Venezuelan refugees, on a scale from one to ten. Altruism large business owners is the reported support for policies to assist large business owners who struggled during the pandemic. Self-reported altruism is each participant's reported level of altruism on a scale from one to ten. ICW Index stands for an index from the inverse covariance matrix (Anderson 2008) constructed with outcomes in columns 1, 2, 5, and 7. Round denotes the round of fieldwork in which the outcome in the column heading was collected. Heteroskedastic-robust standard errors are in parentheses.

experimental and survey measures of prosociality toward members of different groups as well as other dimensions of individuals' beliefs and attitudes. The indicator variable $Ancestry_i$ assumes the value of one for participants who had an ancestor confined to the leper colony, and $\mathbf{X}_i$ constitutes a vector of individual-level controls that include marital status, dummies for primary and secondary attainment, gender, age, age squared, and number of children.

Table 3 presents the results for different dimensions of altruism as captured by the dictator games described in Section III and various survey measures. Notably, descendants of the excluded tend to be significantly more altruistic toward both in- and out-groups. Column 1 shows that they allocated Col\$2,680 more, on average, to in-group members (57.6 percent more with respect to the mean in-group allocation).

Moreover, as shown in column 2, they allocated Col\$1,401 more, on average, to out-group members (33.4 percent more with respect to the mean out-group allocation). These altruistic behaviors in dictator games mask an important tendency from descendants of the excluded to favor members of their own group, that is, a tendency toward parochial altruism (Bernhard, Fisbacher, and Fer 2006). Indeed, column 4 examines the significance of the difference between in-group and out-group allocations, showing that descendants of the excluded tend to disproportionately favor members of their own group.

Is the behavior in dictator games consistent with a tendency by descendants of the excluded to behave altruistically toward other outsiders? To address this question, I asked participants about the extent to which they agreed with policies supporting Venezuelan migrants with taxpayer money, which has been a controversial issue in Colombia given the large-scale immigration from this country. I also asked them how much they supported using taxpayer money to support large business owners who had struggled during the recent pandemic. This second out-group is considered privileged and disproportionately powerful in the Colombian context. Columns 5 and 6 in Table 3 show that participants with an excluded ancestor tend to display significantly higher support for these policies, suggesting a higher altruism toward other out-groups. Hence, those with excluded ancestors are more likely to display solidarity toward outsiders who have struggled regardless of whether these outsiders were initially in a state of vulnerability (as Venezuelan migrants) or privilege (as large business owners).

All in all, these results suggest that descendants of the excluded are more likely to behave altruistically overall despite a proclivity to display a greater altruism toward in-groups. This pattern is reinforced by column 7, which indicates that these descendants tend to self-report a higher altruism (on a one-to-ten scale) than those whose ancestors were not excluded. Moreover, column 8 corroborates this pattern, showing that descendants of the excluded have a significantly higher latent altruism, as captured by a summary index based on an inverse covariance matrix (ICW index) with the experimental and the survey outcomes of the first round of fieldwork (Anderson 2008).

Does social exclusion carry a legacy for other dimensions of prosociality? Table 4 examines various survey outcomes to assess this possibility. Column 1 shows that descendants of the excluded have significantly higher trust toward locals than those whose ancestors were not excluded. Conversely, columns 2 and 3 indicate that compared to descendants of the nonexcluded, descendants of the excluded do not exhibit significantly different levels of trust toward Colombians or trust in the Colombian government. These findings are important for two reasons. First, the results in column 1 suggest that descendants of the excluded display stronger prosocial attitudes toward locals than descendants of the nonexcluded (i.e., not only a stronger local altruism, as documented in Table 3, but also local trust).

Second, the results in columns 2 and 3 suggest that these descendants blame neither Colombians nor the government for what happened to their ancestors, a finding that aligns with the historical discussion in Section I. Even though the government was behind the operation, and despite a policy predicated upon an enduring stigma, doctors were the ones who rationalized, implemented, and even physically enforced the policy of isolation. Moreover, as noted in Section I, they were the only

TABLE 4—DIFFERENCES IN PROSOCIALITY AND OTHER ATTITUDES BETWEEN STUDY PARTICIPANTS WITH AND WITHOUT AN EXCLUDED ANCESTOR

	Trust in...						Other attitudes	
	Locals (1)	Colombians (2)	Colombian government (3)	Physicians (4)	HPV vaccine (5)	COVID vaccine (6)	Anti- parasitic take-up (7)	Religiosity (8)
<i>Panel A. No covariates</i>								
<i>Ancestry</i>	0.755 (0.334)	−0.006 (0.306)	−0.077 (0.282)	−1.274 (0.225)	−1.877 (0.434)	−0.260 (0.291)	−0.158 (0.073)	0.253 (0.173)
Observations	275	275	275	439	220	273	174	265
R^2	0.018	0.000	0.000	0.070	0.082	0.003	0.027	0.007
<i>Panel B. With baseline covariates</i>								
<i>Ancestry</i>	0.698 (0.341)	0.054 (0.310)	0.007 (0.285)	−1.348 (0.230)	−1.769 (0.458)	−0.179 (0.289)	−0.147 (0.076)	0.271 (0.197)
Observations	275	275	275	439	220	273	174	265
R^2	0.051	0.016	0.027	0.099	0.112	0.115	0.064	0.055
Mean dep. var.	6.269	5.153	2.764	7.239	6.645	6.777	0.362	9.521
Round 1				✓	✓			✓
Round 2				✓			✓	
Round 3	✓	✓	✓			✓		

Notes: The table documents the legacy of ancestral social exclusion on various dimensions of trust and other attitudes. Each panel-column reports results from a single OLS regression of the dependent variable described in the column heading on the variable(s) described in the row heading(s). Panel B shows the results of regressions that control for gender, age, age squared, number of children, marital status, and indicators for primary and secondary education. *Ancestry* is an indicator that equals one if the participant reports having an ancestor who lived in the colony during the epoch of isolation. Columns 1–6 have as outcomes the level of trust reported by each study participant toward the actor or vaccine described in the column heading (on a one-to-ten scale). HPV refers to the human papillomavirus. Column 7 has as an outcome an indicator that equals one if the study participant in the second round of fieldwork redeemed an antiparasitic offered to them for free. Column 8 has a self-reported level of religiosity (on a one-to-ten scale) as an outcome. Round denotes the round of fieldwork in which the outcome in the column heading was collected. Observations from the second round of fieldwork correspond to those collected without an intervention (i.e., observations from the second round of fieldwork in this table are restricted to the control group—see online Appendix A.34 for further details about the protocol in the second round). Heteroskedastic-robust standard errors in parentheses.

representatives of the state within the colony, which made them the only visible face of the traumatic policy. Hence, from the lived experience of those who were relegated to the leper colony, doctors constitute the salient exclusionary group (as opposed to politicians or Colombians overall).

To examine if there is an adverse legacy on prosociality and attitudes toward the group that was visibly exclusionary, I measured attitudes toward doctors and medicine in various ways. First, I asked participants how much they trusted physicians on a one-to-ten scale. Second, I leveraged the context in Colombia, where there was widespread disinformation regarding the safety of the human papillomavirus (HPV) vaccine, as covered by national and local media. I thus enquired how much participants trusted the safety of the HPV vaccine on a one-to-ten scale. Third, given the hesitancy about COVID-19 vaccines observed in various regions of the world, I asked participants how much they trusted the safety of these vaccines (again, on a one-to-ten scale). Last, I leveraged the region's context, where doctors recommend

that individuals take antiparasitics with some frequency given the risk of drinking nonpotable water. As a part of the second round of fieldwork, I gave all participants vouchers that they could redeem in the local pharmacy in exchange for a full dose of antiparasitics. To the extent that participants would find such treatment more useful if they trusted the recommendations from physicians, the take-up of antiparasitics provides a revealed preference measure of trust in medicine.

Results in columns 4–7 suggest that the medical mistreatment of patients resulted in durable lower trust in doctors and medicine more generally. Specifically, column 4 indicates that descendants of the excluded report significantly lower trust in physicians, a finding consistent with worse attitudes toward the HPV vaccine (column 5) and a significantly lower take-up of antiparasitics (column 7). Although the reported trust in the COVID-19 vaccine is slightly lower for descendants of the excluded (column 6), this difference is not statistically significant.⁶ Overall, however, these descendants tend to display significantly lower levels of trust in physicians and modern medicine, and this mistrust is specific to those who took part in excluding leprosy sufferers. This is consistent with findings from other contexts, which show that historical medical malpractice had a deleterious, durable effect on trust in medicine without carrying a similar legacy for trust in local or national institutions (Lowe and Montero 2021b; Martínez-Bravo and Stegmann 2022).

I next examine whether these outcomes are consistent with administrative health data. Even though vaccination rates are not available at the individual level along excluded ancestry lines, one can examine municipality rates of infant vaccination for different types of vaccines in the past 20 years. Infant vaccines are freely provided by the Colombian state and are thus more likely to capture differences in vaccine hesitancy across municipalities. Considering that nearly three-fourths of descendants of the excluded continue to live in Agua de Dios, differences in vaccination data between Agua de Dios and its contiguous municipalities provide an indirect (albeit imperfect) measure of differences in health behaviors that could stem from the exclusion of leprosy sufferers.

Consistent with the outcomes collected in different rounds of fieldwork, online Appendix A.13 shows significantly lower rates of infant vaccinations in Agua de Dios compared to (i) contiguous municipalities and (ii) distant municipalities similar to Agua de Dios along climatic factors.⁷ This legacy does not align with self-reported health outcomes in the 2005 census: As I document in online Appendix A.14, descendants of the excluded report better health than descendants of the nonexcluded. Despite the limitations of these data (outcomes were self-reported and not cross-validated with medical records), it is possible that the observed mistrust in medicine does not perfectly map onto suboptimal health behaviors across all dimensions.

Finally, I examine if interactions with outsiders within the leper colony could have affected beliefs and behaviors associated with such outsiders regardless of whether they were seen as exclusionary. For instance, interactions between the excluded and

⁶This finding may reflect the massive information campaigns that local and national governments have undertaken to disseminate accurate reports about the vaccine in Colombia, a policy that has been shown to be highly effective in the context of Latin America even for vaccine-hesitant individuals (Argote et. al. 2021).

⁷Online Appendix A.4 presents the geographical location of these distant, climatically-similar municipalities, while online Appendix A.8 provides further details on how these municipalities were selected.

members of religious orders inside the colony could have affected overall attitudes toward religion. Column 8 shows that the differences in religiosity across ancestry status are not statistically or economically relevant, suggesting that the legacy of mistrust in physicians is not capturing an overall effect on beliefs related to those present in the colony. Instead, distinct attitudes are displayed precisely toward those who acted as an exclusionary group in the past.

A. Interpretation of Findings

The patterns of prosociality documented in the results can be understood in light of models that characterize the evolution of local altruism in the presence of conflict (Bowles 2006; Choi and Bowles 2007). These models show that parochial altruism may arise as an evolutionary response that improves the odds of group survival when intergroup conflict is frequent and prevalent.

Although these models are mostly about conflict, their insights shed light on the empirical patterns described above: the mistreatment, stigmatization, and segregation experienced by leprosy sufferers constituted prolonged external attacks akin to prevalent and frequent conflict. These conflict-like circumstances favored the joint development of in-group altruism and hostility toward the out-group exerting such violence. This helps explain the tendency to favor in-groups more than out-groups, the strong local trust, and the mistrust in physicians documented in Tables 3 and 4. Despite not being victims of violent conflict strictly defined, descendants of the excluded exhibit a behavioral pattern that harmonizes with findings from studies on conflict, which show that experiences of violence lead to mistrust toward those seen as adversaries (Petersen 2001; Wood 2003; Rohner, Thoenig, and Zilibotti 2018).

Considering these evolutionary explanations for the results, why should one also expect higher altruism toward some outsiders? One plausible explanation is the tendency of descendants of the excluded to identify with members of other groups that have lived through similar experiences or circumstances—either facing a similar environment or having undergone a trauma or hardship. This tendency to sympathize with others based on their lived hardship may be particularly salient for descendants of leprosy sufferers: these descendants are bound to their in-groups through the knowledge that they also dealt with a legacy of suffering and pain. The knowledge that an out-group has experienced similar circumstances or suffering could thus be enough for these descendants to extend their local altruism to such outsiders.

Table 3 shows that descendants of the excluded are more supportive of Venezuelan refugees in Colombia (another clearly vulnerable group) and are significantly more supportive of large business owners who struggled during the pandemic. Hence, these descendants tend to be sympathetic to outsiders who have suffered regardless of how well-off these outsiders were before the trauma. Moreover, these descendants behave altruistically toward out-groups who have experienced circumstances similar to theirs, even if little more is known about them. Indeed, Table 3 suggests that descendants of the excluded are altruistic toward anonymous inhabitants of a municipality that was historically very similar and relatively proximate to the region of study. This out-group is not a minority or a group that is particularly vulnerable in the Colombian context, suggesting that a stronger prosociality toward them is based not on their minority status but rather

on the extent to which this out-group has endured circumstances similar to those of descendants of the excluded.

These findings are of broader relevance for several reasons. First, differences in prosociality toward in- and out-groups have been identified as one of the underlying behavioral features explaining contemporary cleavages in ideology (Enke, Rodriguez-Padilla, and Zimmermann 2022a, b). These results suggest that historical social exclusion may thus be a key contributor to political polarization. Second, descendants of the excluded continue to display mistrust toward the historically exclusionary group even when the mistreatment has ceased to exist for generations, suggesting that development policies associated with an exclusionary group may be undermined by historical incidences of exclusion. Third, the fact that the behavioral scars from social exclusion are akin to those from conflict suggests that policies aimed at tackling and mending the scars of violent conflict should be contemplated for historical instances of social exclusion.

B. Threats to Identification and Robustness Checks

This subsection examines the identification assumptions that are embedded in equation (1) and shows evidence suggesting that violations to such assumptions are unlikely.

Pre-existing Differences.—One concern is that the historical nonexcluded population in the region of study was significantly different than those who were secluded in the colony. Despite the absence of rich microdata from the epoch of isolation, the evidence discussed in Section II suggests that the scope for such characteristics as an explanation for the patterns is limited. Online Appendix A.6 examines the two variables available in the census of 1918, male and female population, to examine this possibility. Moreover, it combines the historical census data with information collected by the colony's chief doctor about each patient's place of origin circa 1920. This allows me to compute the population density and male share that patients were exposed to, on average, before arriving in the colony.

To the extent possible given data constraints, online Appendix A.6 reveals that patients in the colony and people living in surrounding municipalities exhibited similar gender ratios and population densities. Additionally, patients came from places that, on average, were also similar to the municipalities surrounding the colony along those lines. Patients' municipalities of origin—located in the central region of Colombia, which is also where the local nonexcluded population lived—ensure there were no pre-existing differences along ethnic, racial, religious, or linguistic lines. Moreover, as noted by the colony's chief doctor, patients were mostly agricultural peasants, which was the same occupation held by those in surrounding areas. These similarities are reinforced by the results discussed in Table 1, which examines the first microdata available from the region of study and shows that the population that was likely secluded exhibited attributes that were not significantly different from those of the historical, local nonexcluded population. All in all, these results make it unlikely that pre-existing differences between the population of patients in the colony and the local nonexcluded population can explain the baseline patterns.

Selection in Out-Migration.—One concern is that selective out-migration from the region of study could drive some of the results. Several results mitigate this concern: First, I examine the 2005 census, which contains information on the migration status of individuals and the reason behind their decision to migrate. This allows me to assess the reasons behind the migration of those who out-migrated from the former leper colony and those of out-migrants from the surrounding region. Notably, the distribution of migration reasons is almost identical for these two groups (online Appendix A.15), and there are no significant differences in the share of migrants from each place who reported each of the reasons for out-migrating recorded in the census (online Appendix A.16).

Second, online Appendix A.17 uses data from the World Values Survey to show that in Colombia there is no association between altruism and the two main reasons that motivated the out-migration of people from the region of study (employment and family). Third, online Appendix A.18 uses data from the 2005 census to compare the observable attributes of those who emigrated from the former colony and those of émigrés from the municipalities surrounding it. The table shows no discernible differences between these out-migrants along the observable traits reported in the census. Combined, these results reinforce the claim that selected out-migration is an unlikely explanation for the baseline patterns.

Selection in In-Migration.—A related concern stems from selective in-migration; some participants in the control group could be descendants of people who migrated to the region of study, and these in-migrants could in turn be distinctively unaltruistic or pro-medicine. I assess the likelihood of this concern with several pieces of information. First, descendants of the nonexcluded in the sample allocated 21 percent of their endowment in the dictator games, a figure remarkably close to the behavior displayed by Colombians in other dictator games (Cardenas and Carpenter 2008). Second, I collected survey data from distant municipalities that are climatically similar to the region of study as well as from an economically depressed neighborhood in Bogotá.⁸ Notably, as online Appendix A.19 shows, descendants of the nonexcluded in the main sample report levels of altruism and trust in medicine comparable to those of subjects in these distant places. On the other hand, descendants of the excluded report a significantly higher altruism and a significantly lower trust in medicine relative to this alternative control group. All in all, descendants of the nonexcluded are not distinctively unaltruistic or pro-medicine, suggesting that the baseline patterns are unlikely to be spuriously driven by selected in-migration to the region of study.

Social Desirability.—Another possibility is that descendants of the excluded tend to be more prone to social desirability or social conformity. In the context of the study, such an inclination could lead to a willingness to comply with the expectations of the research team. To examine this possibility, I asked participants how much they cared about the opinions that superiors or their neighbors have about them. Online Appendix A.20 shows that these descendants do not exhibit a

⁸Online Appendix A.4 maps these distant, climatically similar municipalities, while online Appendix A.8 discusses and presents the climatic dissimilarity index used in selecting these municipalities.

significantly higher concern for their neighbors' opinions. They are also significantly less likely to care about the opinion of superiors, which contrasts the concern that they are more willing to comply with the expectations and opinions of researchers (who could be viewed as holding power in the experimental setting). The table also examines reported trust in researchers and shows no significant differences in this dimension across ancestry status. If anything, descendants of the excluded tend to report a slightly lower trust in researchers, which would make them less likely to trust that researchers would honor their allocations to anonymous recipients. These results suggest that participants' responses are not driven by a willingness to comply with the research teams' expectations.

Income and Education.—Table 1 shows no historical imbalance in income or educational attainment across excluded status. However, differences in these dimensions could have materialized after the policy of isolation (e.g., via a persistent stigma), which could explain the patterns through channels different than the ancestral experiences of exclusion. To examine this possibility further, I collected data that serve as a proxy for the income group to which participants belong.⁹ Online Appendix A.21 suggests that the share of participants who belong to each income group is not economically or statistically different across ancestral lines. Moreover, online Appendix A.22 examines the heterogeneity of outcomes based on income, suggesting that social exclusion's legacy on prosociality is not specific to an income group.

As with income, several results suggest that higher human capital investment is an unlikely explanation for the baseline results. First, the results discussed in Tables 3 and 4 are economically and statistically the same when accounting for educational attainment. Second, as presented in online Appendix A.17, data from the World Values Survey suggest that the association between secondary educational attainment and altruism is neither economically nor statistically significant in Colombia. Third, online Appendix A.23 examines the heterogeneity of the main results based on educational attainment and shows that ancestral social exclusion robustly predicts the prosocial outcomes for those who did and those who did not complete secondary education.

Place-Based Patterns.—I next examine if the results are driven by those who reside in the former colony. Online Appendix A.24 presents the average outcomes for participants living in and outside the municipality that used to be the colony (Agua de Dios), split by ancestral status. The figure suggests that the comparison of participants with and without excluded ancestors yields consistent results regardless of place of residence. Moreover, online Appendix A.25 presents the results of regressions accounting for municipality-of-residence fixed effects as well as the results of regressions examining heterogeneity based on municipality of residence. Reassuringly, the table shows that comparisons of participants with and without

⁹Specifically, in the third round of fieldwork, I collected the estrato of each participant, a categorization of households that serve as a proxy for income, which ranges from one (lowest socioeconomic status) to six (highest socioeconomic status). Individuals in Colombia are keenly aware of their estrato. The heterogeneities that I examine in online Appendix A.22 are thus restricted to the outcomes that I collected in the third round of fieldwork.

excluded ancestors are immune to accounting for municipality-of-residence fixed effects. As for heterogeneity based on place of residence, the results pertaining to altruism appear to be stronger for (but not entirely concentrated on) descendants of the excluded who reside in the former leper colony, although I lack power to detect a meaningful place-based pattern. Overall, these results suggest that local conditions in the former colony may accentuate (but not fully explain) some behavioral attributes among descendants of the excluded (e.g., local altruism).

C. External Validity

As discussed in Section I, the exclusion endured by those with leprosy in Colombia is consistent with the mistreatment suffered by individuals with the disease in other regions of the world. Still, the patterns described above might stem from the specificities of the exclusion endured by leprosy sufferers in Colombia. Online Appendix A.26 uses data from the Standard Cross-Cultural Sample (SCCS) to examine the consistency of these findings with patterns observed across ethnicities. For a subset of 186 ethnicities across the globe, the SCCS contains data on whether different pathogens, including leprosy, are present and widespread. These data can be combined with information about the extent to which parents in each ethnicity inculcate different behaviors in children. The results in the table suggest that ethnicities in which leprosy is widespread and severe are more likely to instill into children trust in community members.¹⁰ The table also shows that the prevalence of other diseases do not predict these patterns and that ethnicities where leprosy is severe and widespread are not more likely to instill other behavioral attributes. These findings are consistent with the stronger local prosociality encountered in descendants of leprosy sufferers in Colombia.

A related concern is whether the baseline patterns are specific to leprosy or if they indeed reflect a broader behavioral pattern deriving from social exclusion. To address this question, I examine data from second-generation migrants in the European Social Survey, which asks respondents if they belong to a demographic discriminated against. Online Appendix A.27 shows that second-generation migrants who have been discriminated against are significantly more likely to display altruism and parochialism than those who have not. Conversely, second-generation migrants who have been discriminated against are not more likely to display other dimensions of behavior. Overall, these conditional correlations suggest that the baseline results largely reflect a behavioral pattern deriving from social exclusion, broadly defined.

V. Mechanisms

Thus far, I have shown that descendants of the excluded tend to be parochially prosocial, extend their local altruism to outsiders who have experienced hardship, and tend to mistrust or be wary of those who actively participated in their ancestors' exclusion. The results along ancestral lines are consistent with the transmission of

¹⁰The SCCS codes the variable "trust" in a way that captures the relevance of "confidence in social relationships, especially toward community members outside the family" (Barry et al. 1976, p. 95), which is consistent with a legacy of stronger local prosociality.

such cultural attributes from one generation to the next. However, several questions remain unanswered. What specific cultural praxis is at play, and how does it contribute to the behaviors one encounters among descendants of the excluded?

This section presents evidence suggesting that historical narratives from excluded ancestors, in the form of stories and facts, help explain why their descendants display the behaviors described in Section IV. There are several reasons why these narratives could help explain these behaviors. Ancestral stories and facts about the colony's history may shape and activate memory, evoking past experiences and bringing to mind norms of behavior (Bordalo, Gennaioli, and Shleifer 2020). Moreover, following Bénabou, Falk, and Tirole (2020), the content recurrently transmitted through these narratives may justify actions based on morality. The themes discussed in historical narratives from excluded ancestors may thus raise the saliency of social exclusion for their descendants, cuing them into behaving in a way consistent with their ancestors' experiences. The scope for these behavioral effects may be particularly relevant in the former leper colony of Agua de Dios, where those who endured the exclusion often share their lived traumas via oral histories (Botero-Jaramillo, Mora-Blanco, and Quesada-Jimenez 2017).

To examine how individuals react to historical narratives from the excluded, I leverage the knowledge of locals about histories that circulate frequently in the region of study, in line with approaches that have been recently employed by scholars in the development economics literature (Wantchekon, Klasnja, and Novta 2015; Sánchez de la Sierra 2020). Specifically, in the summers of 2018 and 2019 I held several conversations with a local oral historian about the stories and facts that leprosy sufferers shared about their lived experiences.¹¹ Based on my exchanges with this historian, I designed a randomization exercise to examine the scope of these narratives in triggering the baseline behaviors discussed in Section IV. Between May and June 2019, I recruited 360 participants in the region of study, following the same procedure described in Section III and collecting outcome variables similar to those in the first round of fieldwork. The key difference was that participants in this round were randomly exposed to different historical narratives.

Before answering the survey, participants in the randomization exercise were randomly assigned to one (and only one) of the following groups: participants who answered the survey with no additional information (pure control), participants who were exposed to a locally salient narrative that highlighted the stigma and mistreatment endured by leprosy sufferers (ExclusionHist), participants exposed to a locally salient narrative that highlighted the historical misconceptions pertaining to leprosy that were prevalent among medical practitioners (MedicineHist), and participants exposed to a story about local trees in the region that is orthogonal to the historical traumatic experience of those excluded in the colony (FloraHist).

These treatments are akin to priming different contextual elements that may trigger various dimensions of behavior, following the tradition of several papers in psychology and economics (Bargh, Chen, and Burrows 1996; Shih, Pittinsky, and Ambady 1999; Shariff and Norenzayan 2007; Benjamin, Choi, and Strickland 2010; Cohn, Maréchal, and Noll 2015). They are distinct in that the prime is based on the

¹¹ I am indebted to Efraín Oyaga for his valuable help and the time he spent talking with me about the history of Agua de Dios.

content of historical narratives that circulate locally and reflect the experiences of excluded ancestors, allowing me to examine how narratives from the excluded may influence their descendants' behavior.

Online Appendix A.28 shows that the observable attributes of participants in the randomization exercise are balanced across treatment arms, suggesting that the randomization of these interventions was successful. Moreover, online Appendix A.29 shows that participants exposed to the falsification treatment (information about local trees) did not display significantly different outcomes when compared to the pure control group. Hence, the control group in the results that follow combines participants subjected to this falsification treatment and those in the "pure control" (no intervention) group.

Figure 3 depicts the unconditional averages of the outcomes and 90 percent confidence intervals for each subgroup of participants with the same ancestral status and belonging to the same treatment arm. Moreover, online Appendix A.30 shows the distributions of the outcome variables for each of these subgroups. These figures show that across treatments, those who have excluded ancestors exhibit higher in-group altruism, stronger parochial altruism, and a lower average trust in physicians. However, two notable differences emerge with respect to the first round of fieldwork. First, those without an excluded ancestor also display some in-group favoritism. As noted in Section III, this second out-group was nearly three times as far from the region of study as the original out-group, which may have increased the salience of its out-group status for all participants in this round of fieldwork.

A second and more important difference is that there are no significant differences in out-group altruism between descendants of the excluded and descendants of the nonexcluded. The out-group in this round was selected with the goal of providing other outsiders who were also historically and socioeconomically similar to people in the region of study. The fact that the increased out-group altruism does not replicate with outsiders in the second round thus implies that objective similarity of experiences may not be enough for descendants of the excluded to extend their local altruism to an outsider, suggesting that perceived similarity is what drives the extension of altruism to some out-groups.

To formally examine the effect of these historical narratives on the behavior of study participants, I estimate the following expression:

$$\begin{aligned}
 (2) \quad Y_i = & \alpha + \beta \mathbf{1}\{ExclusionHist = 1\}_i \\
 & + \theta \mathbf{1}\{MedicineHist = 1\}_i + \mu Ancestry_i \\
 & + v(Ancestry_i \times \mathbf{1}\{ExclusionHist = 1\}_i) \\
 & + \omega(Ancestry_i \times \mathbf{1}\{MedicineHist = 1\}_i) + \vartheta' \mathbf{X}_i + \epsilon_i,
 \end{aligned}$$

where *ExclusionHist* and *MedicineHist* are the treatment groups described above; *Ancestry_i* is an indicator that equals one if respondent *i* has an excluded ancestry; *Y_i* is one of the outcomes described above; and *X_i* is a vector of demographic characteristics of respondents including gender, marital status, age, number of children, and dummies for primary and secondary educational attainment.

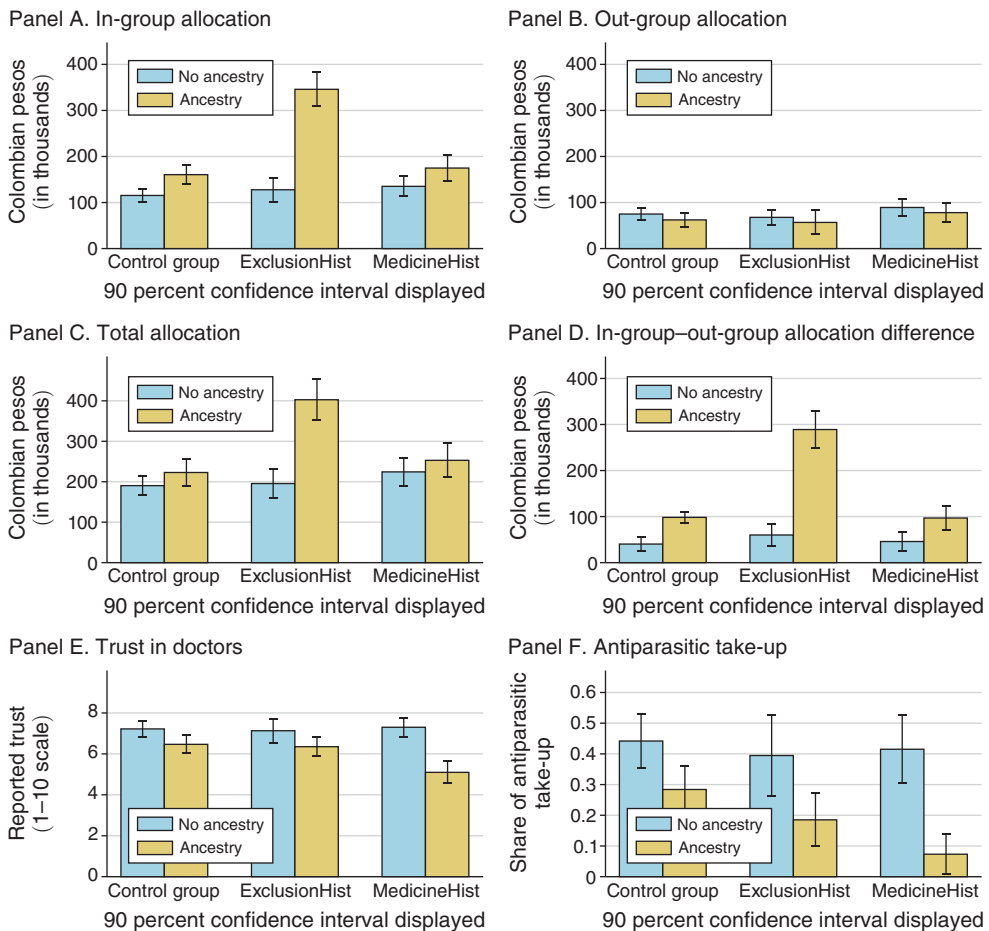


FIGURE 3. BEHAVIOR OF STUDY PARTICIPANTS AFTER HEARING DIFFERENT HISTORICAL NARRATIVES, SPLIT BY ANCESTRAL STATUS

Notes: The figures depict the average outcomes collected in the second round of fieldwork across different narratives treatments, split by ancestral status. Ancestry is an indicator that equals one for participants who reported having an ancestor who lived in the colony during the epoch of isolation. ExclusionHist corresponds to participants that heard a historical narrative about the stigma and mistreatment endured by those who suffered from leprosy. MedicineHist corresponds to participants that heard a narrative about the historical medical errors regarding leprosy. Control group corresponds to participants in the pure control or those who heard a placebo story about the history of a local tree. Participants could allocate up to Col\$800,000 per round to an anonymous receiver either from the same municipality (panel A) or from a distant municipality (panel B). Panel C examines the total allocation that participants chose to give in these modified dictator games. Panel D displays the difference between participants' allocations to in-group and out-group recipients. Values in panels A, B, C, and D are reported in thousand Colombian pesos of 2019 (US\$1 was approximately Col\$3,350 at the time, and Col\$800,000 amounted to a monthly minimum wage). Trust in physicians (panel E) corresponds to reported trust in physicians on a one-to-ten scale. Antiparasitic take-up (panel F) is an indicator variable that equals one for participants who redeemed a dose of antiparasitics offered for free to every study participant in the second round of fieldwork.

Table 5 shows the results of estimating equation (2). The results corroborate that different historical narratives accentuate the baseline differences displayed by those with an excluded ancestor. Conversely, these changes are not observed in those who have no excluded ancestry. Column 1 indicates that participants with an excluded ancestor who were reminded of the stigma suffered by their ancestors (ExclusionHist)

TABLE 5—EFFECT OF NARRATIVES TREATMENTS ON BEHAVIOR OF STUDY PARTICIPANTS

	In-group allocation (1)	Out-group allocation (2)	Total allocation (3)	In-group– out-group allocation (4)	Anti- parasitic take-up (5)	Trust in physicians (6)
<i>Panel A. No covariates</i>						
<i>Ancestry</i>	45.219 (16.535)	–12.727 (12.912)	32.492 (26.701)	57.947 (12.935)	–0.158 (0.073)	–0.755 (0.388)
<i>ExclusionHist</i> × <i>ancestry</i>	172.943 (33.226)	1.816 (23.836)	174.759 (48.050)	171.127 (32.179)	–0.052 (0.122)	–0.025 (0.619)
<i>MedicineHist</i> × <i>ancestry</i>	–5.473 (28.260)	1.531 (22.099)	–3.943 (44.003)	–7.004 (25.254)	–0.184 (0.109)	–1.449 (0.603)
<i>ExclusionHist</i>	12.414 (19.388)	–7.237 (13.847)	5.177 (28.146)	19.651 (18.522)	–0.047 (0.098)	–0.089 (0.453)
<i>MedicineHist</i>	19.783 (17.129)	14.245 (14.435)	34.029 (27.180)	5.538 (16.272)	–0.027 (0.088)	0.081 (0.392)
Observations	360	360	360	360	360	360
R^2	0.292	0.013	0.133	0.366	0.072	0.076
<i>ExclusionHist</i> × <i>ancestry</i> – <i>MedicineHist</i> × <i>ancestry</i>	178.417 (36.821)	0.285 (26.890)	178.702 (53.096)	178.132 (36.587)	0.132 (0.127)	1.425 (0.668)
<i>Panel B. With baseline covariates</i>						
<i>Ancestry</i>	40.452 (17.156)	–14.821 (13.731)	25.630 (27.728)	55.273 (14.032)	–0.157 (0.074)	–1.015 (0.393)
<i>ExclusionHist</i> × <i>ancestry</i>	179.361 (33.920)	6.660 (25.019)	186.021 (49.646)	172.701 (32.990)	–0.056 (0.121)	0.197 (0.602)
<i>MedicineHist</i> × <i>ancestry</i>	–3.535 (28.882)	3.555 (22.777)	0.019 (45.186)	–7.090 (25.770)	–0.190 (0.110)	–1.271 (0.619)
<i>ExclusionHist</i>	3.570 (20.235)	–10.586 (14.429)	–7.016 (29.608)	14.157 (18.937)	–0.050 (0.097)	–0.330 (0.440)
<i>MedicineHist</i>	13.454 (17.839)	12.860 (15.327)	26.314 (28.714)	0.593 (16.786)	–0.032 (0.090)	–0.067 (0.399)
Observations	360	360	360	360	360	360
R^2	0.319	0.036	0.160	0.389	0.101	0.127
<i>ExclusionHist</i> × <i>ancestry</i> – <i>MedicineHist</i> × <i>ancestry</i>	182.896 (37.148)	3.105 (27.420)	186.002 (54.111)	179.791 (36.546)	0.133 (0.127)	1.468 (0.656)
Mean dep. var.	171.992	70.847	242.839	101.144	0.314	6.667
Round 1						
Round 2	✓	✓	✓	✓	✓	✓
Round 3						

Notes: The table examines the effects of reminding participants of different narratives, mediated by the ancestral status of participants. Each panel-column reports results from a single OLS regression of the dependent variable described in the column heading on the variables described in the row headings. Panel B shows the results of regressions that control for gender, age, age squared, number of children, marital status, and indicators for primary and secondary education. *ExclusionHist* and *MedicineHist* are indicators equal to one if the participant was exposed to a narrative about the stigma suffered by those with leprosy or about the historical medical misinformation about leprosy, respectively. *Ancestry* is an indicator that equals one for participants who reported that they have an ancestor who lived in the colony during the epoch of isolation. Participants entered a lottery for Col\$800,000 and were asked how they would allocate such a prize between themselves, an anonymous person from the same municipality, and an anonymous person from a distant municipality (see online Appendix A.34 for further details on the protocol and the out-group). Outcomes in columns 1 and 2 correspond to the allocations selected for the in-group (column 1) and the out-group (column 2) recipients. Total allocation (column 3) is the sum of these allocations, and in-group–out-group allocation (column 4) corresponds to the difference between these two allocations. Antiparasitic take-up (column 5) is an indicator that equals 1 for participants who redeemed a dose of antiparasitics offered for free to every study participant in the second round of fieldwork. Trust in physicians (column 6) corresponds to reported trust in these professionals on a one-to-ten scale. Round denotes the round of fieldwork in which the outcome in the column heading was collected. Heteroskedastic-robust standard errors in parentheses.

allocated Col\$172,943 more, on average, to in-group recipients than their peers in the control group (an increase of 21.6 percentage points when examined as a share of the amount that each participant could distribute). Conversely, column 2 shows that this difference is statistically and economically insignificant when examining the allocation to out-groups as an outcome.

Columns 3 and 4 show that descendants of the excluded who were exposed to a narrative about the stigma endured by their ancestors display significantly higher altruism, but the increase is entirely directed to in-groups. These results suggest that historical narratives about the suffering of ancestors are effective at triggering local altruism: descendants of the excluded are persuaded to behave altruistically toward those who are also bearers of the intergenerational burden of such trauma.

I next examine if trust in medicine is affected by these historical narratives. Columns 5 and 6 show that descendants of the excluded who were reminded of historical medical misconceptions about leprosy exhibit an 18.4 percentage point decrease in the likelihood of redeeming the antiparasitic voucher relative to their peers in the control group. Moreover, they exhibit a 21.7 percent decrease in reported trust in physicians relative to the mean reported trust. Importantly, online Appendix A.31 shows that these treatments do not have a significant effect on descendants' perceptions of lawyers or engineers. Hence, the content of historical narratives triggers a change in attitudes toward those who were directly involved in the exclusion of their ancestors and not toward other experts or professionals.

The evidence presented in Table 5 also suggests that different historical narratives may be more effective at activating different dimensions of behavior. A formal comparison of the treatments' effects among descendants of the excluded reveals that narratives about the exclusion of ancestors are more effective than narratives about medical misinformation at triggering in-group altruism, whereas the latter is more effective than the former at fomenting wariness toward doctors. Moreover, the fact that descendants react to narratives highlighting the misinformation of doctors suggests that part of the current observed mistrust reflects not only doctors' participation in excluding and mistreating patients but also the fact that they lacked effective treatments in the past.

It is worth noting that the limited sample size reduces the precision of the estimates and only allows me to detect large size effects. With a larger sample size, I could have concluded something about the potential adverse effect of the stigma narrative on health outcomes, which, currently, I cannot detect. Moreover, I did not include a joint treatment that combined both narratives, which prevents me from commenting on the interplay of these different narratives. Despite these limitations, the present exercise is an important step toward understanding how the content of historical narratives shared by ancestors who experienced a historical shock may affect the behavior of their descendants.

As discussed in Section III, I asked participants in the third round of fieldwork (a different set of participants who were surveyed two years after the randomization exercise) to share what they knew about the historical role of physicians in the colony. Based on their open-ended responses, I construct different variables to (i) examine whether the narratives shared by descendants of the excluded are consistent with the treatments I implemented in the second round of fieldwork and (ii) see if this content is salient because it was shared by excluded ancestors. Column 1 of

TABLE 6—DIFFERENCES IN HISTORICAL NARRATIVES OF STUDY PARTICIPANTS WITH AND WITHOUT AN EXCLUDED ANCESTOR

	Reported knowledge of doctor's historical role (1 = reports/exhibits)				Source of historical knowledge (1 = reports source)		
	Historical knowledge (1)	In stigmatizing (2)	Doctors uninformed (3)	Negative sentiment (4)	Direct ancestors (5)	Horizontal or oblique source (6)	Institutional source (7)
<i>Panel A. No covariates</i>							
<i>Ancestry</i>	0.085 (0.037)	0.100 (0.055)	0.114 (0.047)	0.164 (0.065)	0.396 (0.059)	−0.149 (0.049)	−0.044 (0.045)
Observations	275	275	275	242	275	275	275
R^2	0.016	0.013	0.026	0.025	0.156	0.028	0.003
<i>Panel B. With baseline covariates</i>							
<i>Ancestry</i>	0.074 (0.038)	0.096 (0.056)	0.104 (0.049)	0.150 (0.067)	0.425 (0.060)	−0.143 (0.051)	−0.071 (0.047)
Observations	275	275	275	242	275	275	275
R^2	0.080	0.058	0.075	0.074	0.175	0.070	0.084
Mean dep. var.	0.880	0.211	0.127	0.566	0.345	0.236	0.156
Round 1							
Round 2							
Round 3	✓	✓	✓	✓	✓	✓	✓

Notes: The table examines the content and sources of local historical narratives about doctors in the leper colony. Data comes from open-ended responses by participants in the third round of fieldwork. Each panel-column reports results from a single OLS regression of the dependent variable described in the column heading on the variable(s) described in the row heading(s). Panel A shows the results of regressions that do not account for individual-level covariates, while panel B shows the regression results when accounting for gender, age, age squared, number of children, marital status, and indicators for primary and secondary education. *Ancestry* is an indicator that equals one for participants who reported having an ancestor who lived in the colony during the epoch of isolation. Reported knowledge is an indicator that equals one if the participant reported knowing something about the historical role of doctors in the colony. In stigmatizing is an indicator variable that equals one if the participant's response mentions that people in the colony were stigmatized/mistreated by doctors. Doctors uninformed is an indicator variable that equals one if their response mentions that doctors were misinformed about leprosy in the past. Negative sentiment is an indicator that equals one if a sentiment analysis of a participant's response using the "sentimentr" text analysis package by Rinker (2021) predicts a negative sentiment (see online Appendix A.33 for additional details about this sentiment analysis). Columns 5–7 have as outcome variables indicators that equal one if the participant reports that they learned their historical narrative from the source in the column heading. Round denotes the round of fieldwork in which the outcome in the column heading was collected. Heteroskedastic-robust standard errors in parentheses.

Table 6 presents estimates of equation (1) with data from the third round, where the outcome is an indicator variable that equals one if the participant reported any type of knowledge about the historical role of physicians. The column shows that descendants of the excluded are significantly more likely to report knowing something about physicians' historical role. More importantly, I examine indicators for whether the history narrated by each of these participants highlighted the stigma endured by leprosy sufferers (column 2) and the misinformation of doctors (column 3). Notably, historical narratives shared by descendants of the excluded are more likely to mention each of these themes, suggesting these are salient aspects of the historical social exclusion for descendants of those who lived through such trauma.

Online Appendix A.32 examines whether descendants of the excluded are more likely to mention other factually relevant components of the history of exclusion in the colony. In particular, other relevant aspects such as family

separation, isolation, or the role of religious orders are neither frequently nor more likely to be mentioned by descendants of the excluded. This suggests that other dimensions of historical information are not salient despite being crucial aspects of the experience endured by those excluded in the leper colony.

The narratives treatments examined in Table 5 highlight a negative view of physicians, which may or may not coincide with the overall sentiment of the narratives shared by descendants of the excluded. To examine the validity of the narrative treatments' tone, I also examine whether study participants with an excluded ancestor tend to share historical narratives that convey a negative view of physicians in the past. Specifically, I leverage a text analysis package in R named "sentimentr," designed by Tyler Rinker (Rinker 2021). The package calculates the sentiment of a specific response based on the polarity of the words of each response.

Using the package developed by Rinker (2021), I examine whether the narratives shared by each participant conveyed an overall negative sentiment (negative score) or not. Column 4 of Table 6 shows that the responses from descendants of the excluded are significantly more likely to display a negative sentiment than responses from descendants of the nonexcluded. Online Appendix A.33 shows the robustness of this finding to (i) using alternative text analysis packages in R and (ii) using a complimentary methodology that considers words that amplify or attenuate each word's polarity ("not," "however," "but," etc.). Reassuringly, the robustness checks yield consistent results, which align with a lingering mistrust in medicine stemming from both physicians' misinformation in the past and their role as an exclusionary group.

A question that remains is whether descendants of the excluded heard these narratives directly from their ancestors or if they instead learned them from different sources (e.g., formal institutions). To answer this, enumerators asked participants to state the main source from which they learned the historical information they shared. Responses to this question were also open-ended. I categorize each of the sources shared by participants as (i) direct ancestors, (ii) sources that can be either oblique or horizontal (e.g., friends, neighbors, older local people, or relatives other than direct ancestors), (iii) institutional (schools, teachers, universities, museums, government, radio, or hospitals), and (iv) other types of sources that do not fit in these categories (e.g., those who mentioned that they know the history based on their lived experience).

Figure 4 shows the share of participants who report different types of historical sources, split by ancestral status. Notably, more than 60 percent of those who are descendants of the excluded stated they learned these narratives from direct ancestors. Conversely, less than 15 percent mentioned that they learned them from an institution, and less than 15 percent stated a source that can be categorized as either oblique or horizontal. The patterns are starkly different for those who are not descendants of the excluded, with only 20 percent mentioning a direct ancestor and nearly 30 percent mentioning an oblique or horizontal source.

Columns 5–7 in Table 6 formally test the differences between the share of those who mentioned different sources across ancestry status. In line with the patterns observed in Figure 4, column 5 shows that the share of participants who said that direct ancestors were a source is significantly higher for descendants of the excluded. Conversely, as shown in column 6, the share that mentioned a horizontal or oblique

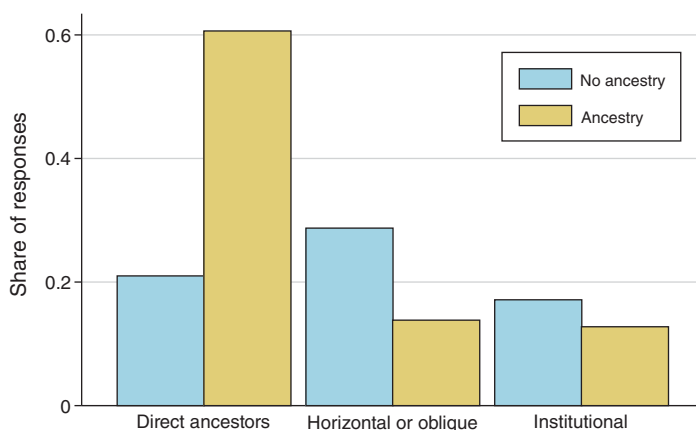


FIGURE 4. HISTORICAL SOURCES OF STUDY PARTICIPANTS WITH AND WITHOUT AN EXCLUDED ANCESTOR

Notes: The figure displays the share of study participants in the third round of fieldwork who reported different types of historical sources, split by ancestral status. Ancestry is an indicator that equals one for participants who reported having an ancestor who lived in the colony during the epoch of isolation. Direct ancestors refers to parents or grandparents. Horizontal or oblique refers to neighbors, friends, close contacts, partners, older local people, and relatives who are not direct ancestors. Institutional refers to schools, universities, museums, government, radio, or hospitals.

source is significantly lower for these descendants. These results suggest that those who personally endured the experience are likely the ones considered as authoritative sources for the stories and facts of what occurred in the colony: descendants of the excluded mostly identify direct ancestors (who personally lived the experience) as a main source, whereas those without an excluded ancestor are more likely to rely on local (noninstitutional) sources outside the family. This interpretation is corroborated by column 7, which shows low levels and insignificant differences when examining the share of those who mentioned institutional sources. New generations are thus informed by historical narratives from those who can provide a direct account of what happened inside the colony instead of relying on second-hand accounts from institutional sources.

One way of interpreting the findings from Tables 5 and 6 is that these patterns reflect a vertical transmission of culture that occurs via historical narratives (Bisin and Verdier 2001). Indeed, descendants of the excluded seem to react to narratives from direct ancestors (but not from sources that could be categorized as horizontal, oblique, or institutional). However, the cultural transmission via historical narratives in the context of this study differs in some respects from Bisin and Verdier's (2001) characterization of vertical transmission of culture. Specifically, the latter is a type of transmission in which parents are altruistic toward descendants' well-being, but they evaluate the actions of their children with their own (parental) preferences. To ensure that children behave in ways that align with parental preferences, ancestors thus undertake costly and deliberate actions to transmit their preferences to their descendants. Vertical transmission through historical narratives would thus imply that excluded ancestors choose what historical experiences to share with their children, that such choice is aimed at increasing the odds of transmitting their traits to descendants, and that sharing such stories is somewhat costly for ancestors.

The results from Table 6 suggest that the role of historical narratives as a cultural vehicle of transmission in the context of this study slightly differs from what Bisin and Verdier (2001) envision as the vertical transmission of culture. Instead of being a deliberate and costly action aimed at transmitting traits to descendants, historical narratives from the excluded tend to be widely shared within and outside the household in the context of this study. Indeed, 85 percent of those without an excluded ancestor claim to know something about what occurred in the leper colony. Moreover, they are as likely as descendants of the excluded to mention factually relevant information about what occurred in the colony (as shown in online Appendix A.32). This suggests that descendants of the nonexcluded are also historically informed about what transpired in the colony. Crucially, they are more likely to be informed by oblique historical sources (i.e., locals outside their family), indicating that they received these histories and facts precisely from locals who were likely excluded. Hence, excluded ancestors likely shared these stories with locals (both descendants and nondescendants), and these contents are well known outside the household, making it unlikely that the narratives are purposeful and costly choices intended only for the transmission of traits to descendants of the excluded.

If both those with and without an excluded ancestor are exposed to narratives from those who were excluded, why does one observe differences in their behavioral response to these narratives? Descendants of the excluded find the content persuasive and salient, which is what columns 1–3 of Table 6 document. Historical narratives from excluded ancestors thus discuss themes that raise the saliency of social exclusion for their descendants. By raising the saliency of social exclusion, the themes discussed in these historical narratives activate behaviors on descendants of the excluded that align with the experiences of their ancestors.

VI. Conclusion

This paper studies the intergenerational consequences of social exclusion on behavior and examines how ancestors' historical narratives help explain these intergenerational patterns. Leprosy, as an affliction that has been globally stigmatized and treated through harsh measures of isolation, provides several empirical advantages as a window into these intergenerational consequences. Two of these crucial advantages are the arguably random nature of the disease in the context of Colombia in the nineteenth and early twentieth centuries and the fact that the disease has nearly vanished for several generations. Such aspects of historical leprosy allow for an examination of the behavioral consequences of social exclusion across generations—even when exclusion is no longer happening.

I have examined outcomes from a lab-in-the-field approach implemented in and around a region previously serving as the site of a leper colony, documenting a complex legacy of prosocial behaviors stemming from ancestral social exclusion. Descendants of the excluded display parochial prosociality in dictator games and survey questions, exhibiting stronger altruism and trust toward in-group members than toward outsiders. The legacy of prosociality toward outsiders depends upon who that outsider is and what they have experienced. On the one hand, descendants of the excluded extend their local altruism to outsiders whom they perceive to have gone through experiences that relate to those of their ancestors, including living

under similar circumstances or experiencing suffering. On the other hand, they tend to display mistrust toward those seen as historically responsible for their ancestors' exclusion. In the context of this study, those with ancestors excluded due to leprosy display mistrust toward physicians, who were discernibly and inextricably involved in the exclusion that occurred in the colony. Conversely, these individuals exhibit no mistrust toward other entities and actors that, despite being factually involved in the exclusion of their ancestors, were not visibly active in doing so (such as politicians and the Colombian state).

The paper also shows that the content of historical narratives from ancestors helps explain these intergenerational patterns of behavior. A randomization exercise reveals that descendants of the excluded display stronger parochial altruism when exposed to locally salient narratives that highlight the suffering of their ancestors. Moreover, they display wariness toward doctors when exposed to a locally salient narrative that highlights the role of these actors in the exclusion endured by their ancestors. The content of these treatments is consistent with the narratives shared by local oral historians as well as those that descendants of the excluded share in their own words. Moreover, these descendants report that excluded ancestors are the main sources of such histories. The fact that descendants of the excluded react to these narratives suggests that the content within them is effective at raising the saliency of exclusion for these descendants, which helps explain why they behave in ways consistent with the experiences of their ancestors despite not being directly exposed to their conditions.

This paper opens several avenues for future research. First, the results suggest that social exclusion is conducive to durable parochialism. An intriguing research avenue is the role that historical social exclusion has played in undermining the cohesion of regions across the globe. Second, the results suggest that political alignment between different excluded groups is probable given their tendency to partially extend their in-group prosociality to outsiders in a state of vulnerability. Future work could explore the role that this behavioral legacy has played in the formation of political coalitions and the circumstances under which it has been conducive to political transformation. Moreover, one could investigate the interplay among competing historical narratives from different groups; for example, by examining the effects of historical narratives from the historical exclusionary group, how they are transmitted, and how they affect the behaviors of different actors, including those of members of the excluded group. Last, the results of this paper show that the legacy of mistrust in the exclusionary group may undermine the success of policies implemented by such a group. A promising area for future research is thus understanding how transitional justice policies may improve the success of development policies associated with groups that have been exclusionary in the past.

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